SESSION 1

Introduction to Databases & Installation

Solved Tasks • SQL Basics • MySQL Setup • MySQL vs PostgreSQL

Learning goal: Understand what databases are, install/connect to MySQL, create databases, and compare MySQL with PostgreSQL.

Session Concepts

1. What is a database?

A database is an organised collection of information. For example, a music-streaming database may store users, songs, artists, playlists, and listening history.

2. What are RDBMS and SQL?

- RDBMS (Relational Database Management System): software that stores related information in tables made of rows and columns, and manages relationships between those tables.
- SQL (Structured Query Language): the language used to create databases and tables, insert and update data, and ask questions about stored data.

3. Database engines mentioned in the session

Database system	Simple description
MySQL	A widely used open-source relational database, often used by websites and transactional applications.
PostgreSQL	An open-source object-relational database known for SQL features, extensibility, and support for complex data workloads.
Oracle Database	A commercial relational database system used in many enterprise environments.
SQL Server	Microsoft's relational database system; commonly managed with SQL Server Management Studio (SSMS).

Why SQL is the backbone of analytics

SQL lets analysts retrieve relevant data, combine tables, clean and filter records, calculate totals and averages, group results, and prepare data for dashboards and reports. Tools such as Excel and Power BI often work with data that can be queried or prepared using SQL.

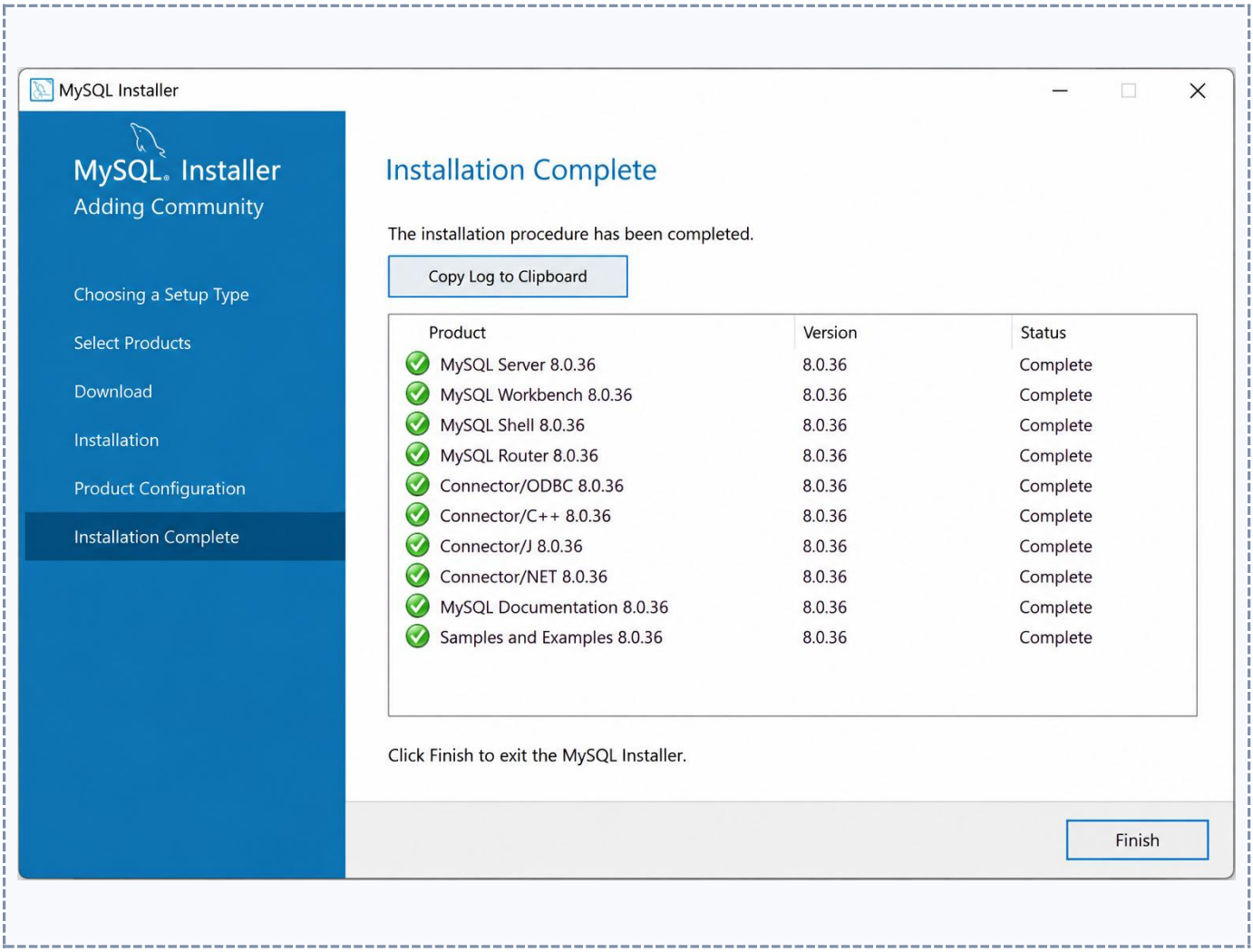
Task 1 — Install MySQL Community Server and capture a screenshot

Answer / steps (Windows):

1. Open the official MySQL Community Server downloads page: https://dev.mysql.com/downloads/mysql/ . Choose the Windows package that matches your system and follow the official installation instructions.

2. Alternatively, for the MySQL 8.0 series, the official MySQL Installer can install MySQL Server and tools such as MySQL Workbench in one wizard: https://dev.mysql.com/downloads/installer/. The official page notes that MySQL Installer is the final installer series for MySQL 8.0; newer server releases use product-specific installers/configurators.
3. Run the installer and follow the prompts. If using MySQL Installer, choose a setup option that includes MySQL Server and MySQL Workbench (for example, Developer Default), if available.
4. Configure the server when prompted. Set and safely remember the root password. Keep the default port unless your setup requires another port.
5. Finish the installation/configuration. Open MySQL Workbench and test the local connection.

Screenshot requirement: You must capture the actual completion window on your own computer. I cannot truthfully provide a screenshot of an installation that has not been performed on your device.



How to take the screenshot: when the installer shows that installation/configuration is complete, press Win + Shift + S, select the completion window, and save or paste the capture here. If the installed version uses a different completion screen, capture the final successful installation/configuration screen.

Task 2 — Connect to MySQL and create music_streaming_db

Use MySQL Workbench (recommended for this task). DB Browser for SQLite is designed for SQLite databases and does not connect directly to a MySQL Server.

1. Open MySQL Workbench.

2. Click your local MySQL connection (often named Local instance MySQL). Enter the root password when requested.

3. Open a new SQL query tab, type the command below, and click the lightning-bolt Execute button.

4. Run SHOW DATABASES; to confirm that the database appears. Then run USE music_streaming_db; to make it the active database.

```
CREATE DATABASE music_streaming_db;

SHOW DATABASES;

USE music_streaming_db;
```

Explanation

- CREATE DATABASE music_streaming_db; creates a new database with that name.
- SHOW DATABASES; lists the databases available on the connected MySQL server.
- USE music_streaming_db; selects the database for subsequent commands in the current session.

Task 3 — Create food_delivery_db

SQL command:

```
CREATE DATABASE food_delivery_db;
```

To verify and select it, run:

```
SHOW DATABASES;
USE food_delivery_db;
```

Explanation: CREATE DATABASE creates the database, SHOW DATABASES verifies that it is listed, and USE selects it as the current database. Execute these statements in MySQL Workbench after connecting to the server.

Task 4 — Three differences between MySQL and PostgreSQL

Comparison	MySQL	PostgreSQL
1. Database model	Commonly described as a relational DBMS. It is widely used for web applications and transactional workloads.	An object-relational DBMS, combining relational tables with additional object-relational capabilities.
2. Data types and extensibility	Supports common numeric, text, date/time, spatial, and JSON types.	Offers a broad set of types such as arrays, ranges, network addresses, and user-extensible types, functions, and operators.
3. Feature emphasis / use cases	Often selected for website backends, e-commerce, and applications where a practical, widely supported relational database is needed.	Often selected for complex queries, specialised data models, data integrity needs, and extensions such as PostGIS for geospatial work.

These are typical differences and use-case tendencies, not strict limits. Both systems can serve many of the same kinds of applications; the better choice depends on the workload, architecture, operational needs, and team experience.

Popular company/app examples

- MySQL: Spotify is listed among MySQL customers on the official MySQL website. See the MySQL Customers page in the references.
- PostgreSQL: Instagram has publicly documented its use of PostgreSQL in its backend/scale-up story. Large companies may use multiple databases, and their architecture can change over time.

Quick Revision

Command / concept	Remember
Database	An organised collection of information.
RDBMS	Manages relational data stored in tables.
SQL	Used to define, query, and manipulate data.
CREATE DATABASE database_name;	Creates a new database.
SHOW DATABASES;	Lists databases on the connected MySQL server.
USE database_name;	Selects the current database.
MySQL Workbench	A graphical tool for connecting to MySQL Server and running SQL.

Official / supporting references

- **MySQL Community Server downloads:** https://dev.mysql.com/downloads/mysql/
- **MySQL Installer downloads and version note:** https://dev.mysql.com/downloads/installer/
- **MySQL Installer Guide:** https://dev.mysql.com/doc/mysql-installer/en/
- **MySQL customers (includes Spotify listing):** https://www.mysql.com/customers/
- **PostgreSQL: What Is PostgreSQL?:** https://www.postgresql.org/docs/current/intro-whatis.html
- **PostgreSQL project overview:** https://www.postgresql.org/about/
- **Instagram and PostgreSQL at scale (EDB article):** https://www.enterprisedb.com/blog/instagram-scales-postgres
- **AWS comparison: MySQL vs PostgreSQL:** https://aws.amazon.com/compare/the-difference-between-mysql-vs-postgresql/